Process Sigma Conversion Table

The Table below shows the relationship between Process Capability Metrics

Long Term Process Sigma, Short Term Process Sigma, Process Yield and DPMO (based on 1 opportunity per unit)

LT Process Sigma	ST Process Sigma	Yield %	DPMO
0.0	**1.5**	**50.00%**	**500000**
0.1	1.6	53.98%	460172
0.2	1.7	57.93%	420740
0.3	1.8	61.79%	382089
0.4	1.9	65.54%	344578
0.5	**2.0**	**69.15%**	**308538**
0.6	2.1	72.57%	274253
0.7	2.2	75.80%	241964
0.8	2.3	78.81%	211855
0.9	2.4	81.59%	184060
1.0	**2.5**	**84.13%**	**158655**
1.1	2.6	86.43%	135666
1.2	2.7	88.49%	115070
1.3	2.8	90.32%	96800
1.4	2.9	91.92%	80757
1.5	**3.0**	**93.32%**	**66807**
1.6	3.1	94.52%	54799
1.7	3.2	95.54%	44565
1.8	3.3	96.41%	35930
1.9	3.4	97.13%	28717
2.0	**3.5**	**97.72%**	**22750**
2.1	3.6	98.21%	17864
2.2	3.7	98.61%	13903
2.3	3.8	98.93%	10724
2.4	3.9	99.18%	8198
2.5	**4.0**	**99.38%**	**6210**
2.6	4.1	99.53%	4661
2.7	4.2	99.65%	3467
2.8	4.3	99.74%	2555
2.9	4.4	99.81%	1866
3.0	**4.5**	**99.87%**	**1350**

LT Process Sigma	ST Process Sigma	Yield %	DPMO
3.1	4.6	99.90%	968
3.2	4.7	99.93%	687
3.3	4.8	99.95%	483
3.4	4.9	99.97%	337
3.5	**5.0**	**99.98%**	**233**
3.6	5.1	99.98%	159
3.7	5.2	99.99%	108
3.8	5.3	99.99%	72
3.9	5.4	99.995%	48
4.0	**5.5**	**99.997%**	**32**
4.1	5.6	99.998%	21
4.2	5.7	99.9987%	13.3
4.3	5.8	99.9991%	8.5
4.4	5.9	99.9995%	5.4
4.5	**6.0**	**99.9997%**	**3.4**
4.6	6.1	99.9998%	2.1
4.7	6.2	99.99987%	1.3
4.8	6.3	99.99992%	0.8
4.9	6.4	99.99995%	0.5
5.0	**6.5**	**99.99997%**	**0.3**
5.1	6.6	99.99998%	0.2
5.2	6.7	99.999990%	0.1
5.3	6.8	99.999994%	0.1
5.4	6.9	99.999997%	0.03
5.5	**7.0**	**99.999998%**	**0.02**
5.6	7.1	99.9999989%	0.01
5.7	7.2	99.9999994%	0.01
5.8	7.3	99.9999997%	0.003
5.9	7.4	99.9999998%	0.002
6.0	**7.5**	**99.9999999%**	**0.001**

Short Term (ST) and Long Term (LT) Process Sigma Explained: Typically Lean Six Sigma projects will run over days, weeks or maybe months (*if they are expected to take longer than 4 months then it may be prudent to re-assess the project scope!*). Because this is 'short' period of time, data collected to assess the process capability has limited power to infer the true process long term capability *[1].

Motorola had empirical evidence showing this to be the case for the vast majority of processes. Thus this shift was built into the Six Sigma methodology. They recognised a shift in process capability of between 1.4 and 1.6 Sigma (average = 1.5 Sigma) from initial assessment when studied over the long term (years +).

Thus a process with a current (Short Term) rating of **6 sigma** will be given a Long Term rating of **4.5 sigma**.

*1 It is likely that other influential variables not recognised in the project work will effect the Long Term process capability after the project work has been complete.